

Section 1 Selecting and Sorting

SQL fundamental: SELECT

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Q1 All columns from the employees table

```
SELECT *
FROM employees;
```

Table employees

id	First name	Last name	department	Salary	hiredate	City
1	Alice	Green	IT	10000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

Q2 All unique departments.

```
SELECT SELECT DISTINCT department
DISTINCT department = 'IT';
FROM employees
```

Department
IT
HR
Finance
Marketing

Q3 first and last names ordered by salary descending.

```
SELECT first-name, last-name
last-name
```

```
FROM employees
```

```
ORDER BY Salary DESCENDING;
```

First-name	Last-name	First-name	Last-name
Alice	Green	Eva	Naidoo
Brian	Lee	Alice	Green
Cathy	Zulu	Cathy	Zulu
David	Mokoena	David	Mokoena
Eva	Naidoo	Brian	Lee

Q4 The top 3 highest-paid employees.

```
SELECT id,  
       first name,  
       last name,  
       salary  
FROM employees  
ORDER BY salary DESCENDING  
LIMIT 3;
```

id	first name	last name	Salary
5	Eva	Maidoo	72 000
1	Alice	Green	70 000
3	Cathy	Zulu	65 000

Section 2 Filtering with WHERE

Q5 employees in the IT department.

```
SELECT id, first name, last name, department  
FROM employees  
WHERE department = 'IT';
```

id	first name	last name	Department Salary
1	Alice	Green	IT 70 000
5	Eva	Maidoo	IT 72 000

Q6 employees in Finance with salary > 60000.

```
SELECT id, first name, last name, department,  
       salary  
FROM employees  
WHERE department = 'Finance'  
AND salary > 60000
```

id	first name	last name	department	Salary
3	Cathy	Zulu	finance	65000

Q7 employees in HR or Marketing.

```
SELECT id, first name, last name, department
FROM employees
WHERE department = 'HR' OR 'Marketing';
```

id	first name	last name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

Q8 employees not in IT

```
SELECT id, first name, last name, department
FROM employees
WHERE NOT department = 'IT';
```

id	first name	last name	department
2	Brian	Lee	HR
3	Cathy	Zulu	finance
4	David	Mokoena	Marketing

Q9 employees in IT, HR, or finance using IN

```
SELECT id, first name, last name, department
FROM employees
WHERE department IN ('IT', 'HR', 'finance');
```

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	finance
5	Eva	Maidoo	IT

Q10 employees in IT with salary > 65000 and City Johannesburg.

SELECT id, first name, last name, department, Salary,
City

FROM employees

WHERE department = 'IT'

AND Salary > 65000

AND City = 'Johannesburg';

id	first_name	last_name	department	Salary	City
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Maidoo	IT	72000	Johannesburg